CS & IT ENGINEERING



Algorithms

Miscellaneous



Lecture No.- 01

By- Aditya sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Heaps

PYQs

Topics to be Covered



- Topic
- Topic
- Topic

TC using Recurrence Tree

About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.





Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic: Miscellaneous

Time Complexity Analysis : (Recurrence Relations)

Approaches	We can find
Back-substitution	Value of Recurrence TC: notation
Master Method / Theorem	TC: notation
Recurrence / Recursion Tree Approach	TC: notation



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Types of TC Recurrences :

1. Symmetric : $T(n)=a\times T(n/b)+f(n), a\geq 1,\ b>1,\ f(n) = +ve$ ⟶ Master mtd

2. Asymmetric

   (a) Case 1 : $T(n)=T(\alpha n)+T((1-\alpha)n)+f(n)$

   (b) Case 2 : $T(n)=T(\alpha n)+T(\beta n)+T(\delta n)\ldots+f(n)$



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Symmetric Recurrence:

$$T(n)=a\times T(n/b)+f(n), a\geq 1,\ b>1,\ f(n) = +ve$$

Size of each sub-problem

Number of sub-problem

Can be solved by:

1. Back substitution
2. Master method
3. Recursion tree approaches

Equation:

$T(n) = 2T(n/2) + 2T(n) = 2T(n/2) + n$



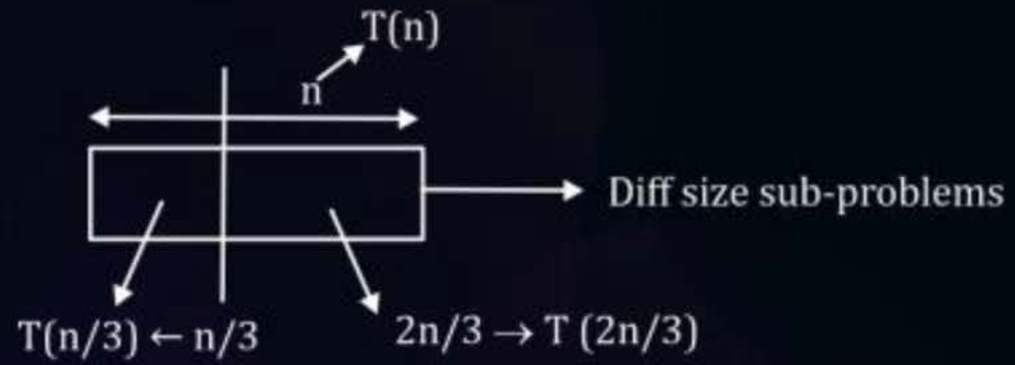
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Asymmetric:

Case-1 $T(n)=T(\alpha n)+T((1-\alpha)n)+f(n)$

$T(n)\ \ 0 \leq \& \leq 1$

Equation:



1. $T(n) = T(n/3) + T(2n/3) + n$
2. $T(n) = T(n/4) + T(3n/4) + n^2$

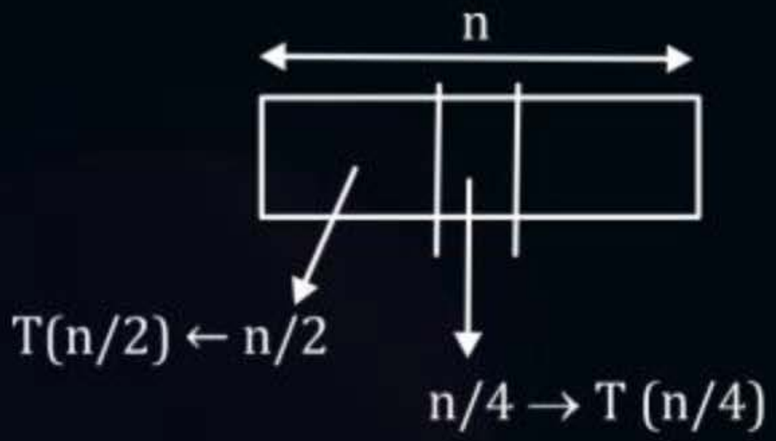
Recurrence Tree appr.



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Asymmetric:

Case-2



$T(n) = T(n/2) + T(n/4) + n^2$

Recursion tree approach.



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Equation:

$T(n) = 2T(n/2) + n$ → Merge sort

1. Back → substitution → value of recurrence?
2. Master method.



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Solution:

Back- substitution:

$T(n) = 2T(n/2) + n$ _____(1)

$T(n/2) = 2T(n/2^2) + n/2$

$T(n) = 2T(n/2^2) + n/2) + n$

$T(n) = 2^2T(n/2^2) + 2n$ _____(2)

$T(n) = 2^3T(n/2^3) + 3n$ _____(3)



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General Term

$T(n) = 2^k T(n/2^k) + k * n$ _____(4)

for B-C , $\frac{n}{2^k} = 1 \Rightarrow n = 2^k$

$\boxed{k = \log_2^n}$

$T(n) = n * T(1) + n * k$

$T(n) = n * c + n \log_2^n$ → value of recurrence

$\boxed{T(n) = O(\log^n)}$ $T(n) = O(n \log_2 n)$



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Master Method:

$T(n) = 2T(n/2) + n$ (given)

$T(n) = a^* T(n/b) + f(n)$

$a = 2$

$b = 2$

~~$F(\infty) = 2$~~

Valid

$\log_b^a = \log_2^2 = 1$

$f(n) = n$

Case 1 :- Is $F(n) = O\left(n^{\log_b a - \varepsilon}\right)$, $\varepsilon > 0$?

$n = O\left(n^{1-\varepsilon}\right)$, some $\varepsilon > 0$?

fails

Case 2 :- If $f(n) = \Theta(n^{\log_b a} * (\log n)^k)$, for some k

a) $k \geq 0$ $\quad n = \Theta(n^1 * (\log n)^k)$

$k = 0$ $\quad n = \Theta(n)$ ✓

Hence, $T(n) = \Theta(n^1 * (\log n)^{0+1})$

$$\boxed{T(n) = \Theta(n \log n)}$$



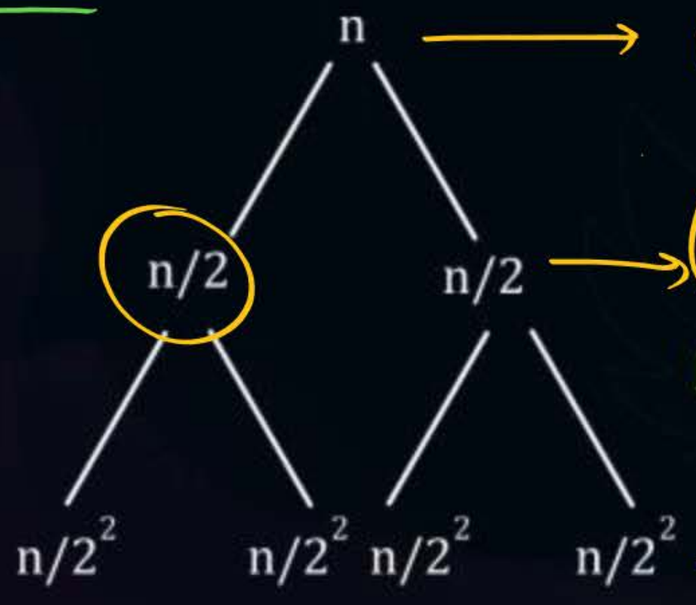
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3

Recurrence tree Approach:

$T(n) = 2T(n/2) + n$

Cost/fee of recursion



n

$(n/2 + n/2 = n)$

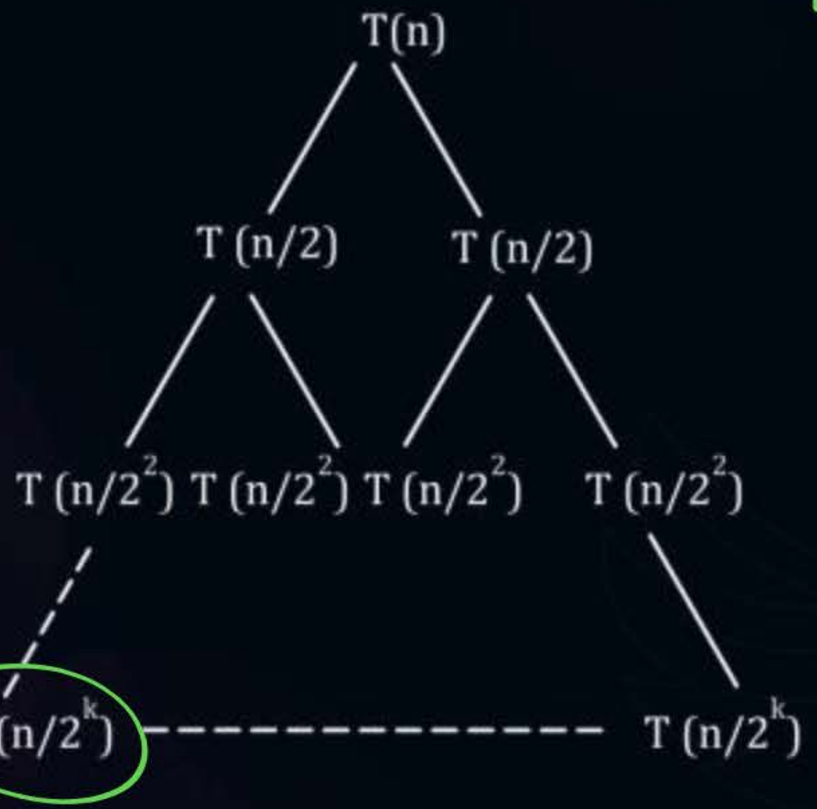
Every level has cost of 'n'.

$4 \times \frac{n}{2^2} = n$



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K levels



K levels

Assume this is last level (Base condtion)



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At last level, it should be base condition

$T(n/2^k)$

$$\frac{n}{2^k} = 1$$

$n = 2^k \Rightarrow k = \log_2 n$

~~$n = \log_2 n$~~



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Level -1 → T(n) : cost = n

Level -2 → T(n/2) : cost = n

Level -3 → $T(n/2^2)$: cost = n

Level –(k+1) → $T(n/2^k)$: cost = n



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Total number of levels = (k+1)

The cost/ time of every level = n

Hence, total cost/ time = n* (k+1)

= n* k+ n

We know

$k = \log_2^n$

= n * $\log_2^n$ + n

$T(n) = \theta(n\log n_2^n)$

$n \log_2 n$

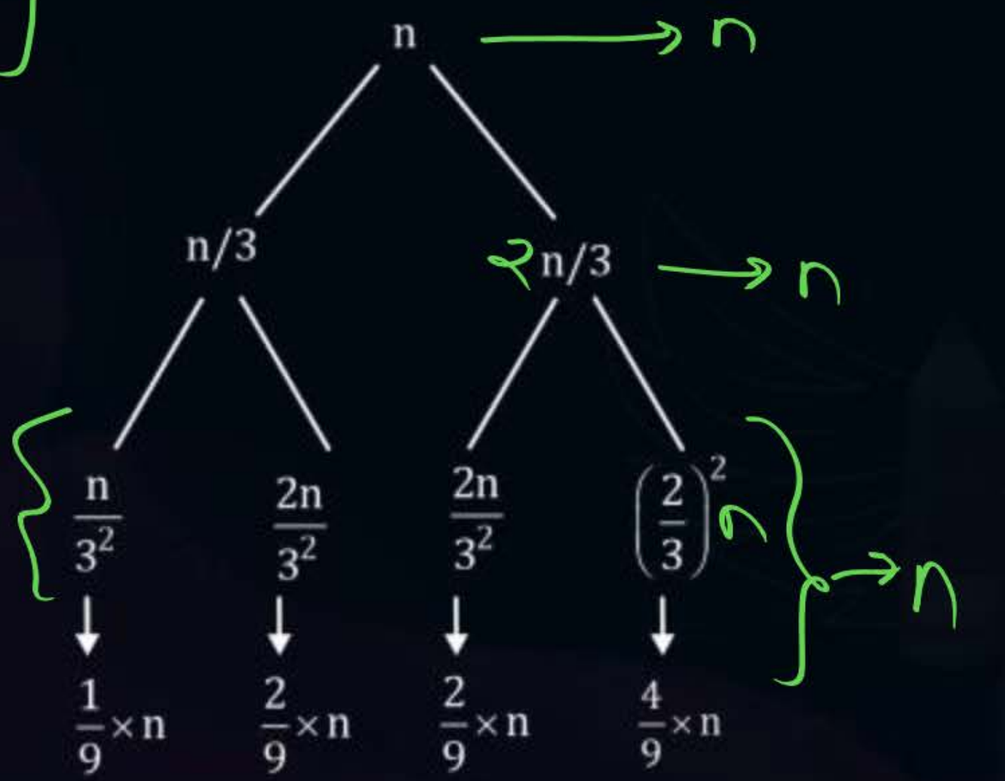
$T(n) = \theta(n \log_2 n)$

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Equation:

$$T(n) = T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + n$$

Asymmetric case -1



$$n \to \left(\frac{1}{9}+\frac{2}{9}+\frac{2}{9}+\frac{4}{9}\right)n$$

$$= \frac{1+2+2+4}{9} \times n$$

$$= \frac{9}{9} \times n$$

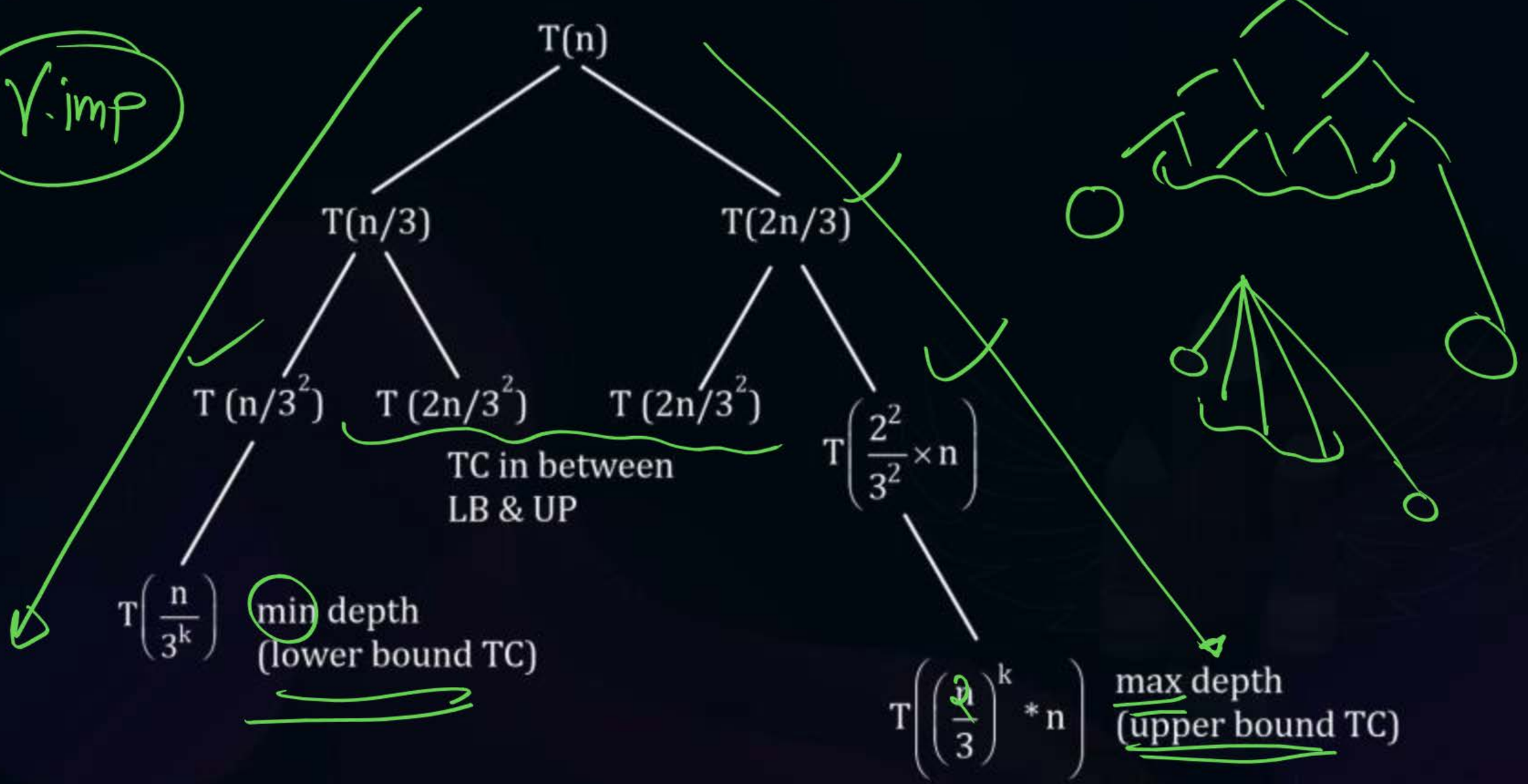
$$= 1 \times n$$

Cost at every level = n

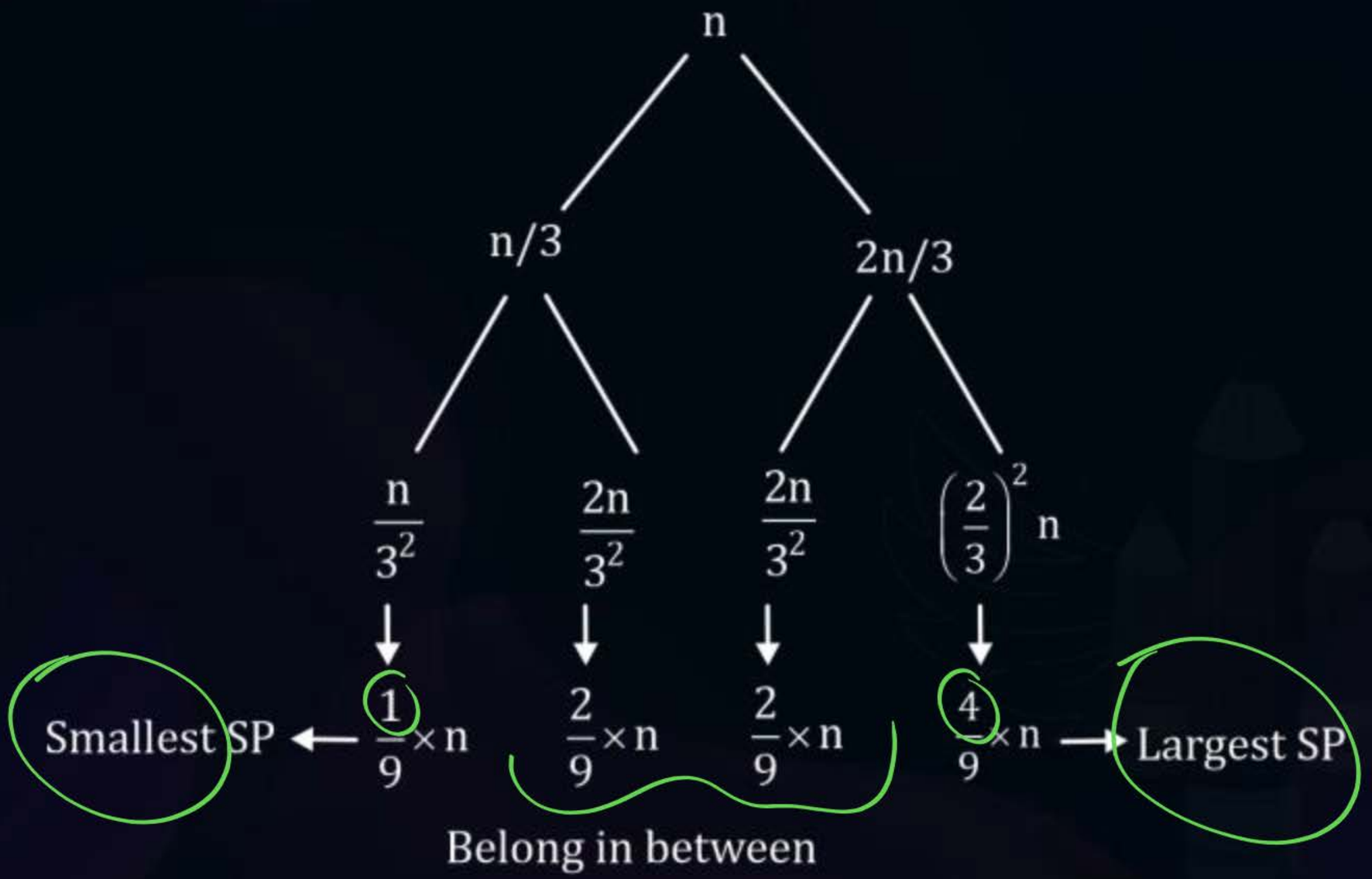
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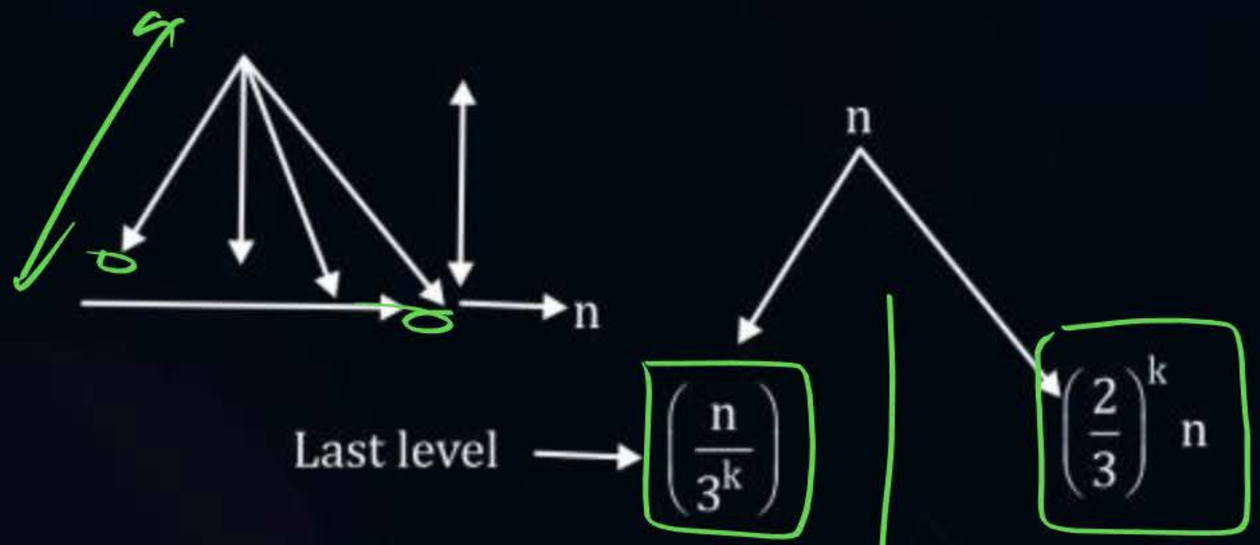
V. imp



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1. For smallest/ min depth (LB of TC)

   At least level k, $\frac{n}{3^k} = 1$ (BC)

   $n = 3^k$

   ~~$k = \log_3^n$~~

   $k = \log_3 n$



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left branch

Assuming there are k levels and cost/time at each level = n

Hence, $T(n) \geq k * n$

$T(n) \geq \log_3^n * n$

$\boxed{T(n) \geq n * \log_3^n}$ _____ (i)

$T(n) = \Omega(\log_3^n * n)$

for UB of TC

$$\left(\frac{2}{3}\right)^k * n = 1 \Rightarrow n = \left(\frac{3}{2}\right)^k \Rightarrow k = \log_{3/2} n$$

TC = Total cost $\leq k * n$

$$\leq (n * \log_{3/2} n)$$

$$TC = O(n \log_{3/2} n)$$



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From (i) & (ii)

$$n * \log_3^n \leq T(n) \leq n * \log_{3/2}^n$$

$T(n) = \Omega(n \log_3(n))$ $T(n) = O(n \log_{3/2} n)$

Approximation $\Rightarrow T(n) = \theta(n \log n)$

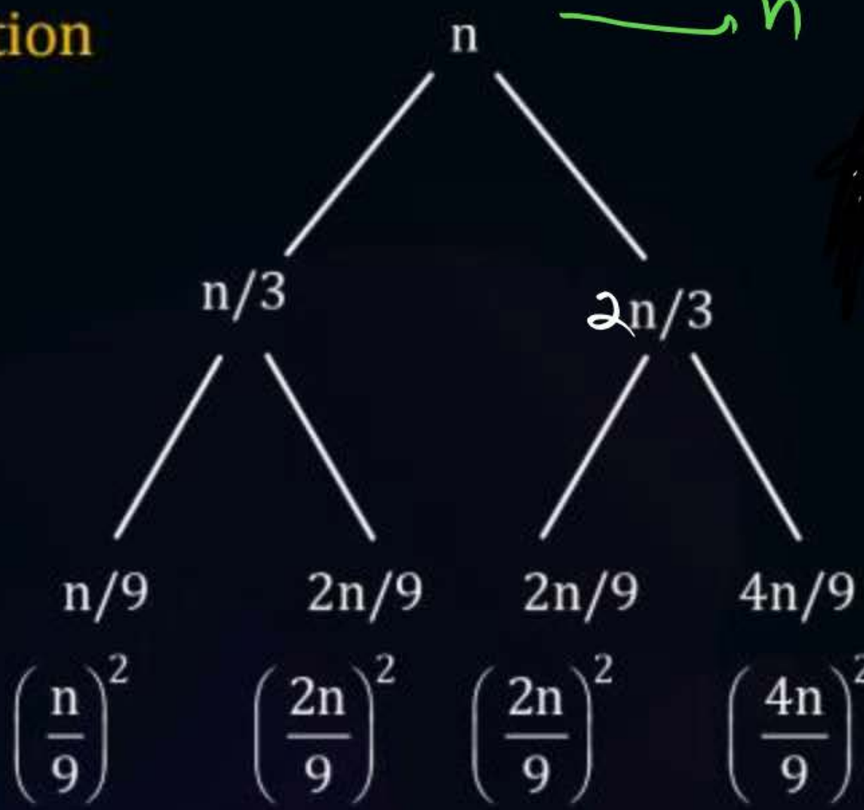
[NAT]

#Q. $T(n) = T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + n^2$

Cost of Recurrence

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Solution



n^2

$$\Rightarrow \left(\frac{n}{3}\right)^2 + \left(\frac{2n}{3}\right)^2 = \frac{n^2}{9} + \frac{4n^2}{9} = \frac{5n^2}{9}$$

$$\frac{n^2}{9^2} + \frac{4n^2}{9^2} + \frac{4n^2}{9^2} + \frac{16n^2}{9^2} = \frac{5n^2}{9^2} n^2$$

$$\left(\frac{5}{9}\right)^2 n^2$$



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Let there be k levels

Total at each level

1st Level $\rightarrow n^2 \rightarrow \left(\frac{5}{9}\right)^0 n^2$

2nd Level $\rightarrow \frac{5}{9}n^2 \rightarrow \left(\frac{5}{9}\right)^1 n^2$

3rd Level $\rightarrow \left(\frac{5}{9}\right)^2 n^2$

k^{th} Level $\rightarrow \left(\frac{5}{9}\right)^{k-1} n^2$



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Hence, Total lost (T(n))

Sum of cost at each level.

$$T(n) = \left(\frac{5}{9}\right)^0 n^2 + \left(\frac{5}{9}\right)^1 n^2 \ldots\ldots \left(\frac{5}{9}\right)^{k-1} n^2$$

$$= \sum_{i=0}^{k-1} \left(\frac{5}{9}\right)^i n^2$$

$r < 1$



$$= \frac{1}{\left(1-\frac{5}{9}\right)} * n^2$$

$$\boxed{\text{Total cost} \approx \frac{9}{4} n^2}$$

$\Rightarrow T(n) : \underline{\theta(n^2)}$



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Shortcut

~~Shortest~~

$n < 1$

$$\sum_{i=0}^{\infty} x^i = \frac{1}{1-x}$$

(approximation)

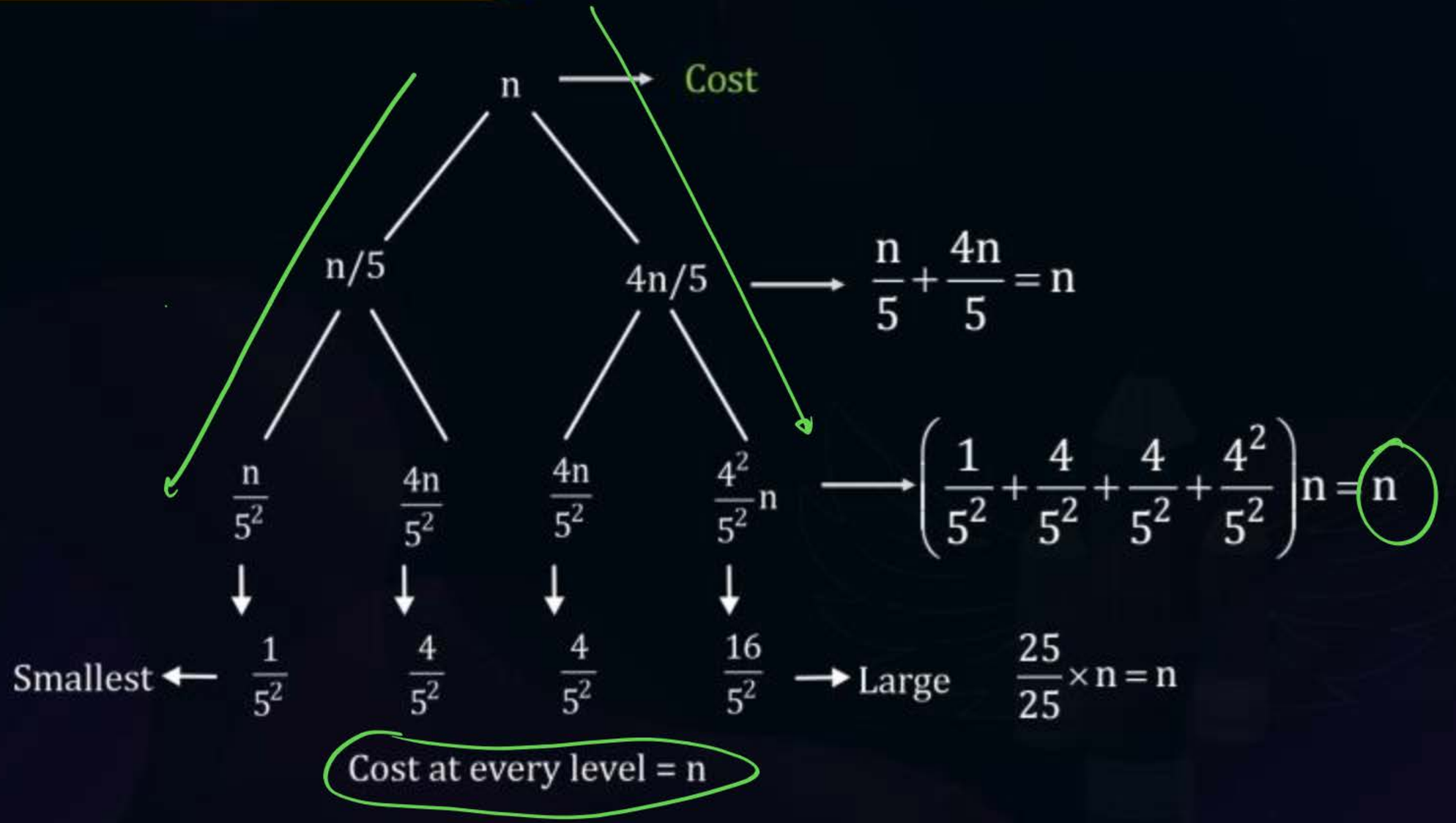
K* cost all each level
here it is difference
Hence, no.

$$T(n) = \theta(n^2)$$

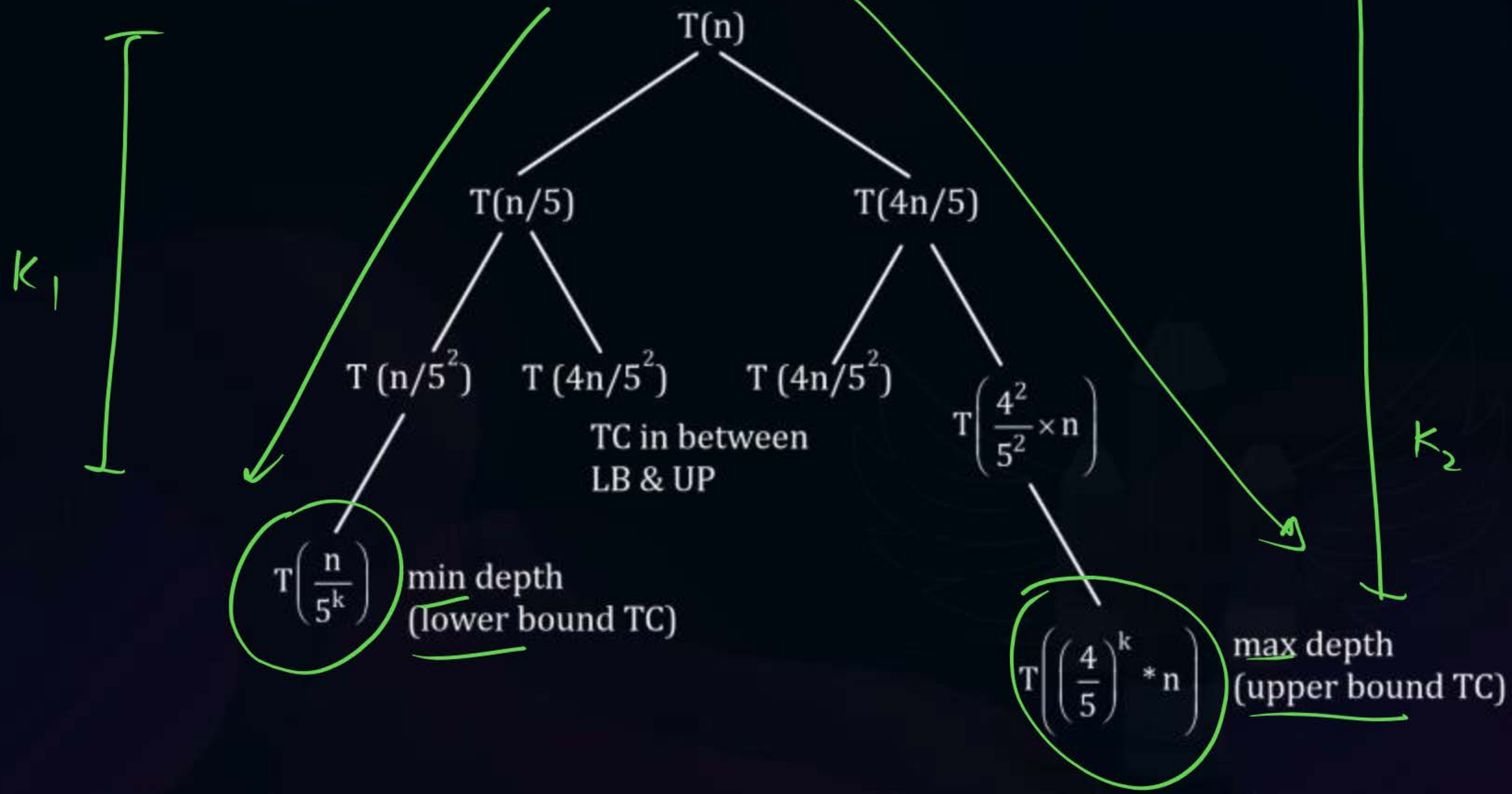
[NAT]

#Q. $T(n) = T(n/5) + T(4n/5) + n$

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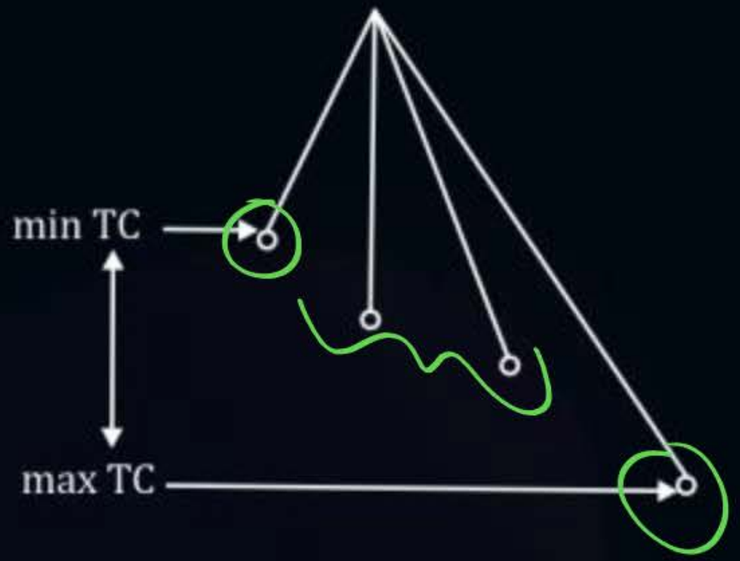


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For

Lowest depth (LB) TC

$\frac{n}{5^k} = 1$

$5^k = n$

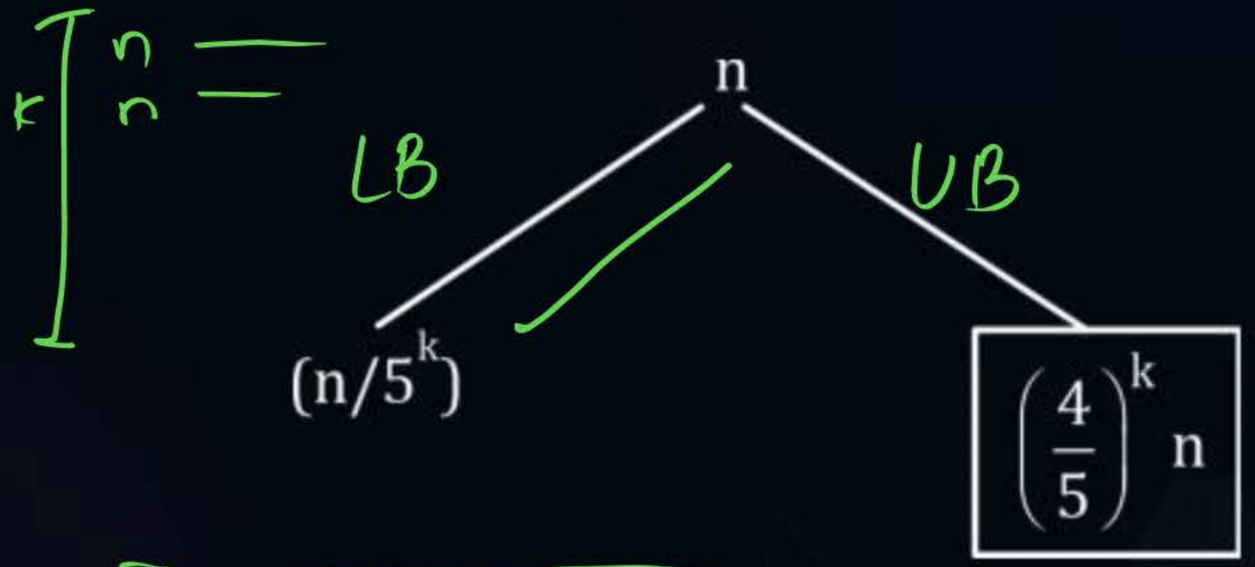
$k = \log_5^n$

Cost at any level = n

$T(n) \geq n*k$

$T(n) > n* \log_5^n$

$T(n) = \Omega(n\log_5^n)$



$T(n) \geq$ Min TC

$T(n) \leq$ Max TC

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For largest depth:

$$\left(\frac{4}{5}\right)^{k} n = 1$$

$$\left(\frac{5}{4}\right)^{k} = n$$

$$k = \log_{5/4} n$$

$$T(n) \leq n * k$$

$$\boxed{T(n) \leq n * \log_{5/4} n}$$



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$$n * \log_5^n < T(n) < n * \log_{5/4}^n$$

$$T(n) = \Omega(n\log_5^n) \qquad T(n) = \Omega(n\log_{5/4}^n)$$

Approximation : $T(n) = \theta\ (n \log n)$

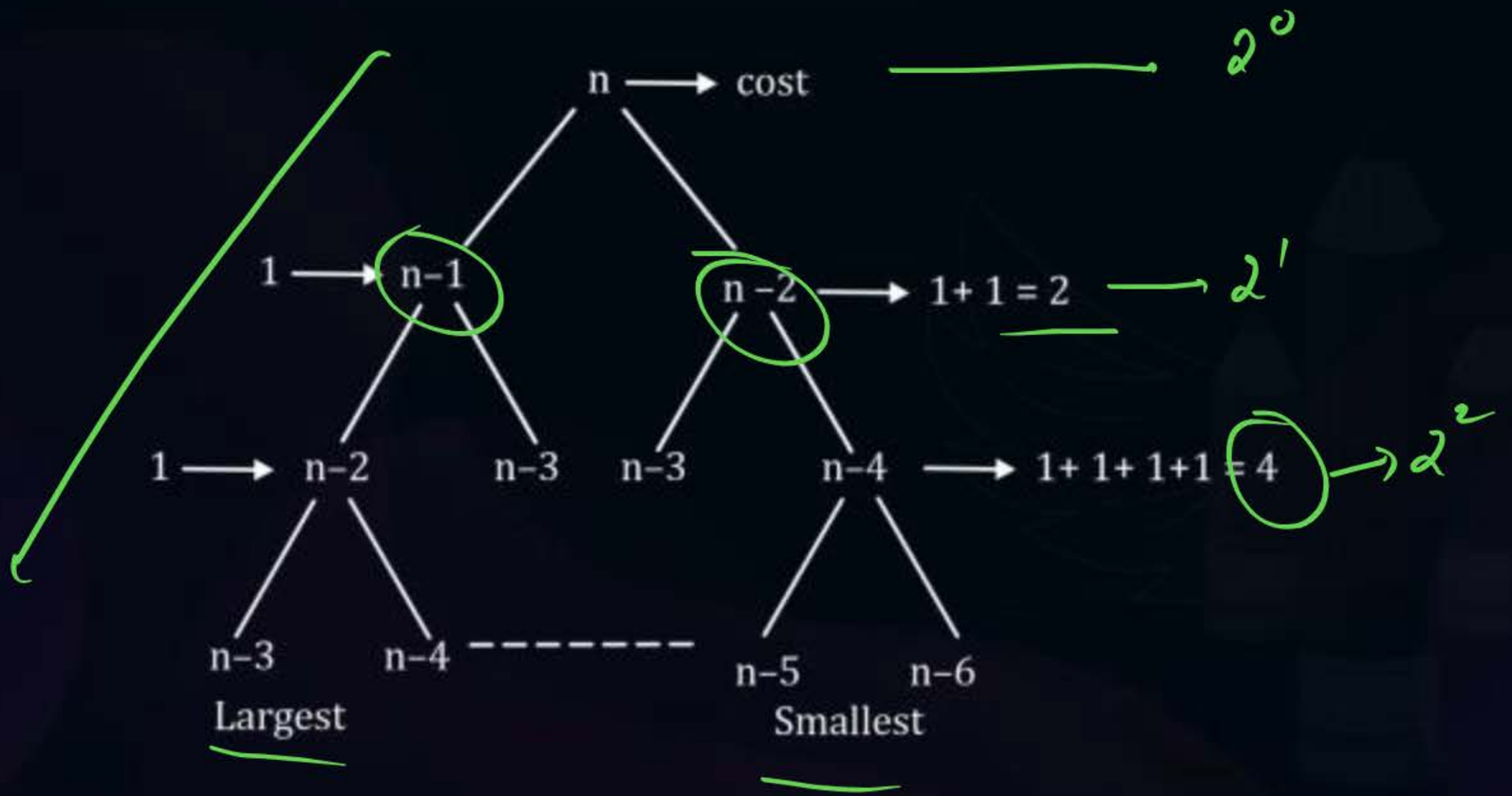
(9) $T(n) = T(n-1) + T(n-2) + 1$



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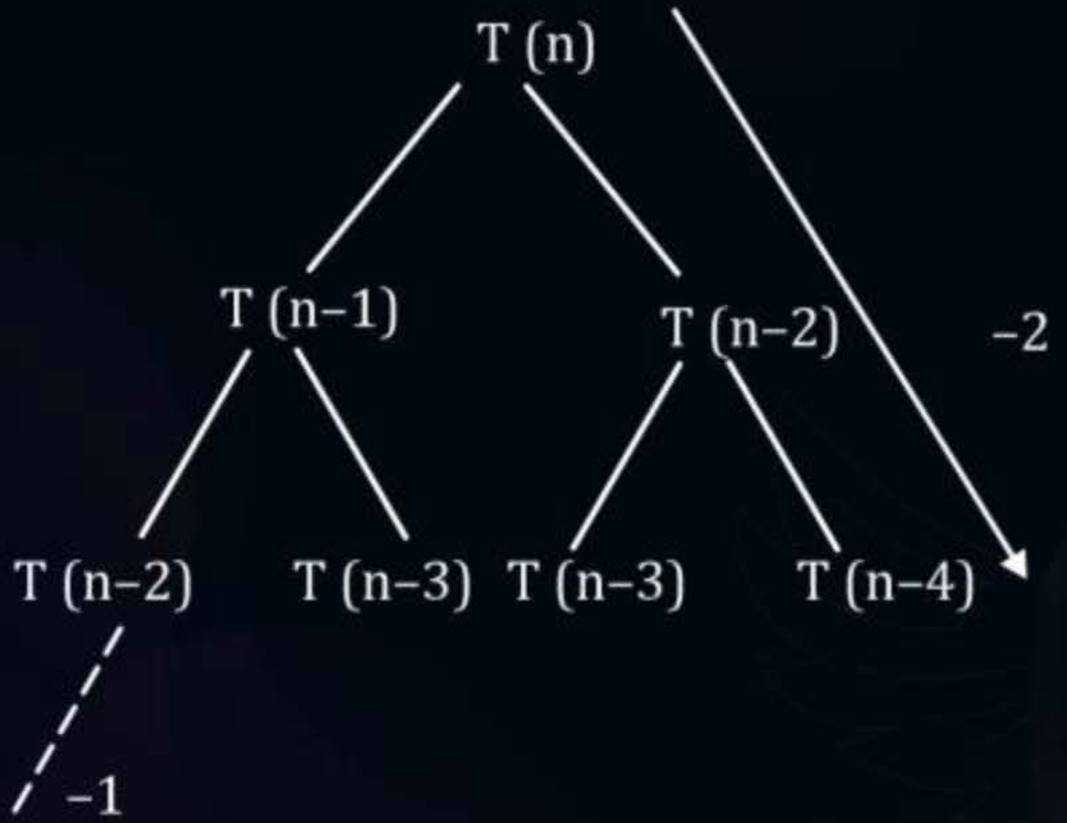
Fibonacci:

TC rearrange $T(n) = T(n-1) + T(n-2) + 1$





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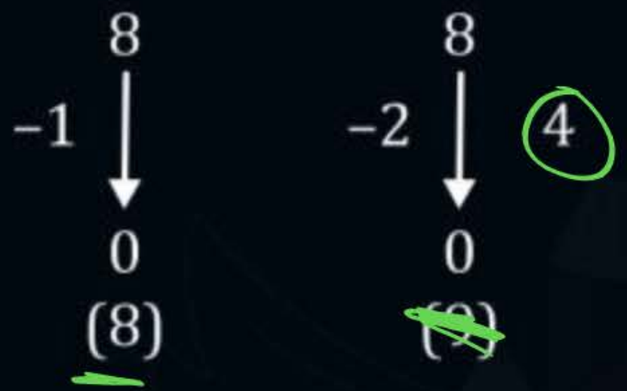


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Left –most branch

$n \rightarrow n-1 \rightarrow n-2 \rightarrow n-3 \ldots\ldots\ldots\ldots\ldots 1$

Number of levels $K = n$ (UB TC)



Right –most branch

$n-2 \rightarrow n-4 \rightarrow n-6 \ldots\ldots\ldots\ldots\ldots$

Number of levels = $n/2$ (LB of TC)

$8 \rightarrow 6 \rightarrow 4 \rightarrow 2 \rightarrow 0$



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Cost of ~~answer~~ any level k

= 20 + 21 + 22 .....2k –1

= (2k –1)

Cost of any branch of k levels

$= 2^0 + 2^1 + 2^2 \cdots 2^{k-1}$

$= 2^k - 1$

1. Left most branch → k =n

$T(n) \leq 2^n - 1$

$T(n) = O(2^n)$

Put k = (n)

Put k = (n/2)



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2. Right most branch → $k = n/2$

$T(n) \geq 2^{n/2} - 1$ $\quad T(n) \geq 2^{n/2} - 1$

$$T(n) = \Omega(2^{n/2})$$



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Conclusion:

Fibonacci

$T(n) = O(2^n)$

~~$T(n) = O(n/2^n)$~~

$T(n) = \Omega(2^{n/2})$

$\neq \Omega(2^n)$

It is $\theta(2^n)$?

↳ Not

Approx

$T(n) = T(n-1) + T(n-2) + 1$

Let $T(n-2) \Rightarrow T(n-1)$

$T(n) = 2T(n-1) + 1$

B.S

[MCQ]

#Q. Consider the following recurrence relation

$$T(n) = \begin{cases} T(n/2) + T(2n/5) + 7n, & \text{if } n > 0 \\ 1, & \text{if } n = 0 \end{cases}$$

which of the following options are correct ?

HW

A $T(n) = \theta(n \log n)$

B $T(n) - \theta(n)$

C $T(n) = \theta(n^{5/2})$

D $T(n) = \theta((\log n)^{5/2})$



2 mins Summary

Recurrence Tree for TC



THANK - YOU